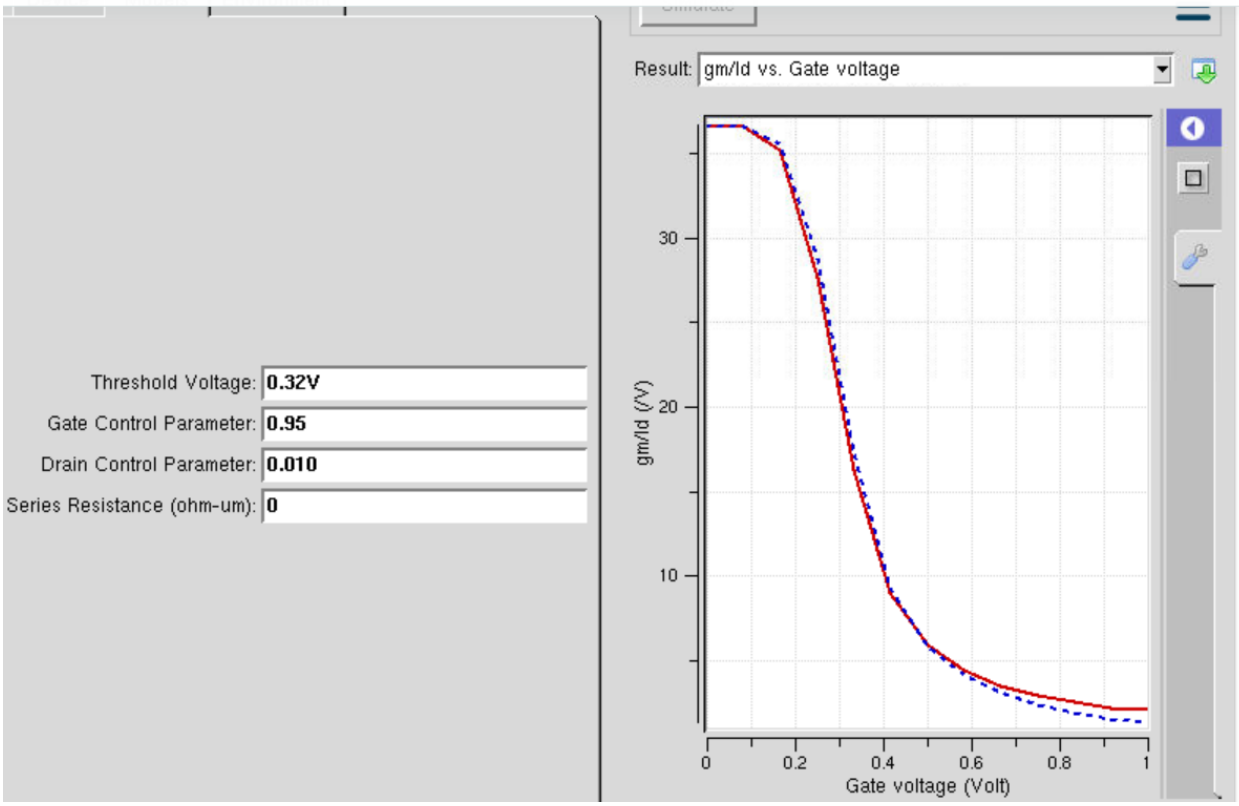
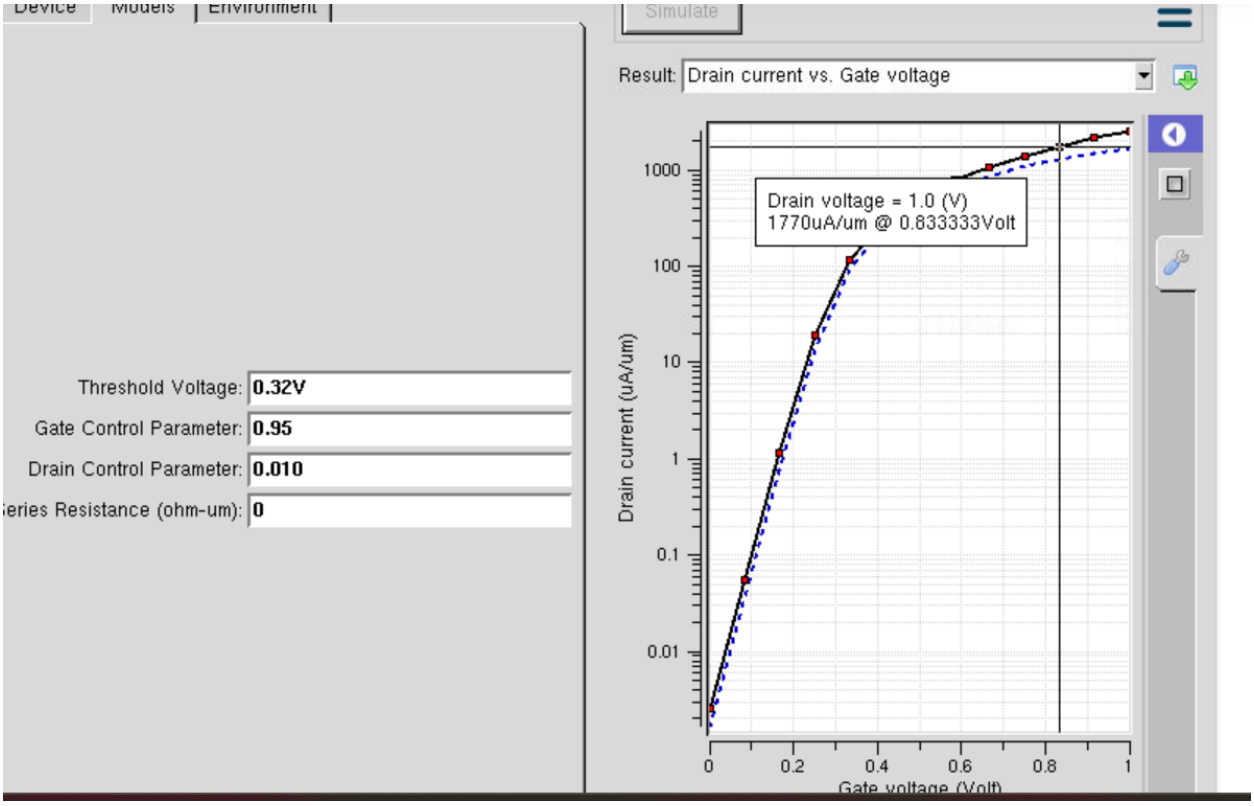
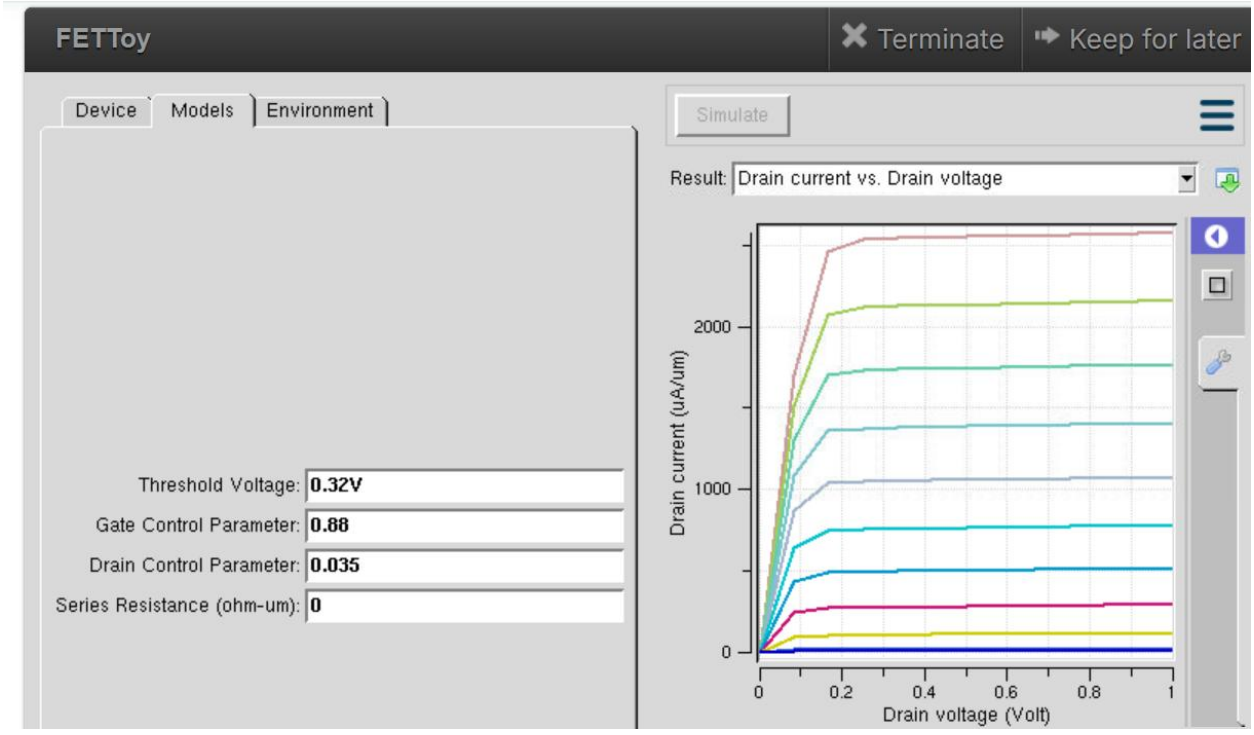
EXPERIMENT – 5
FETToy

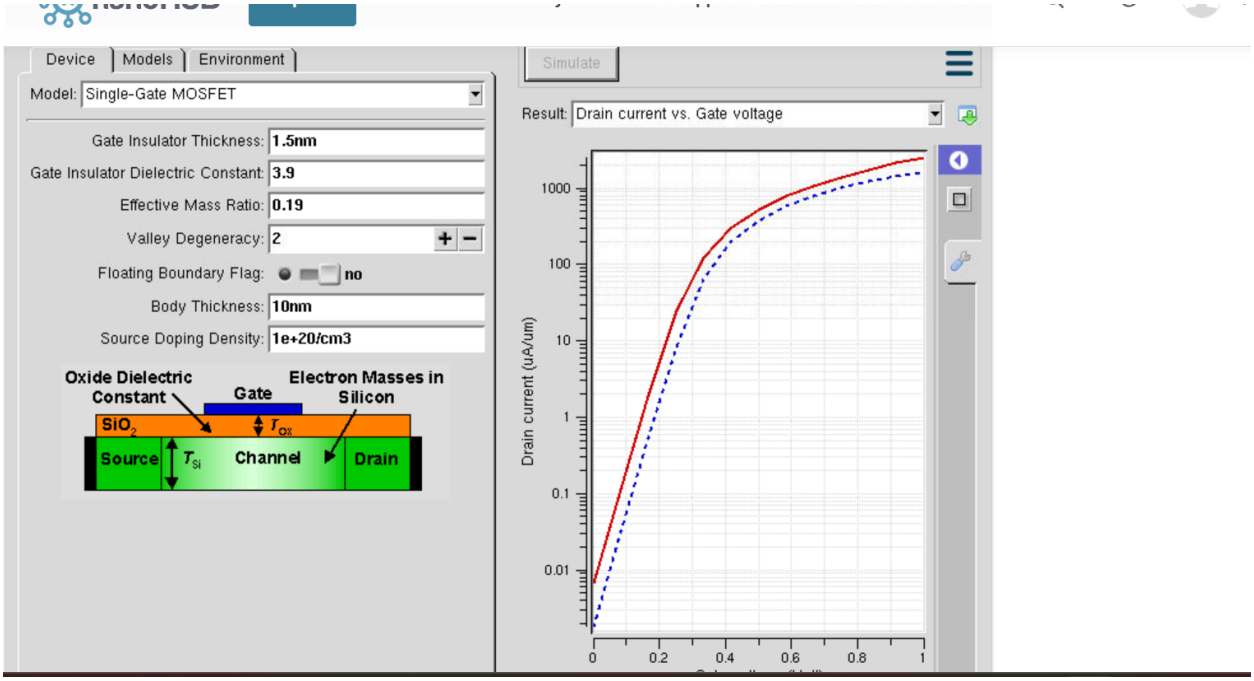
CASE 1

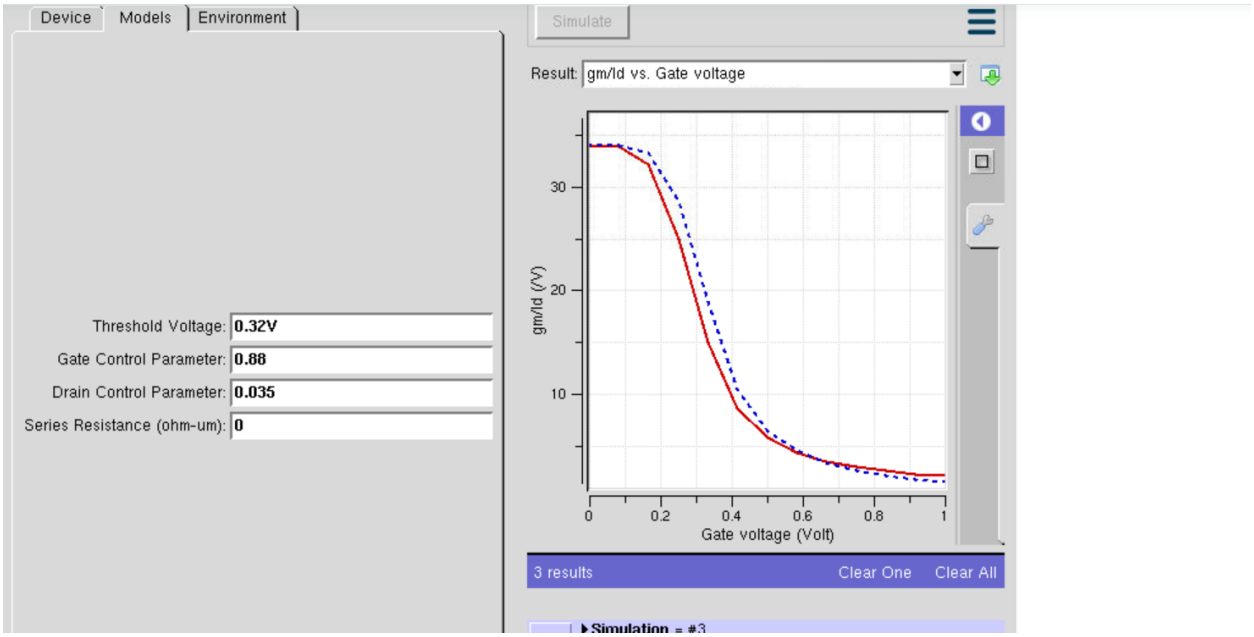




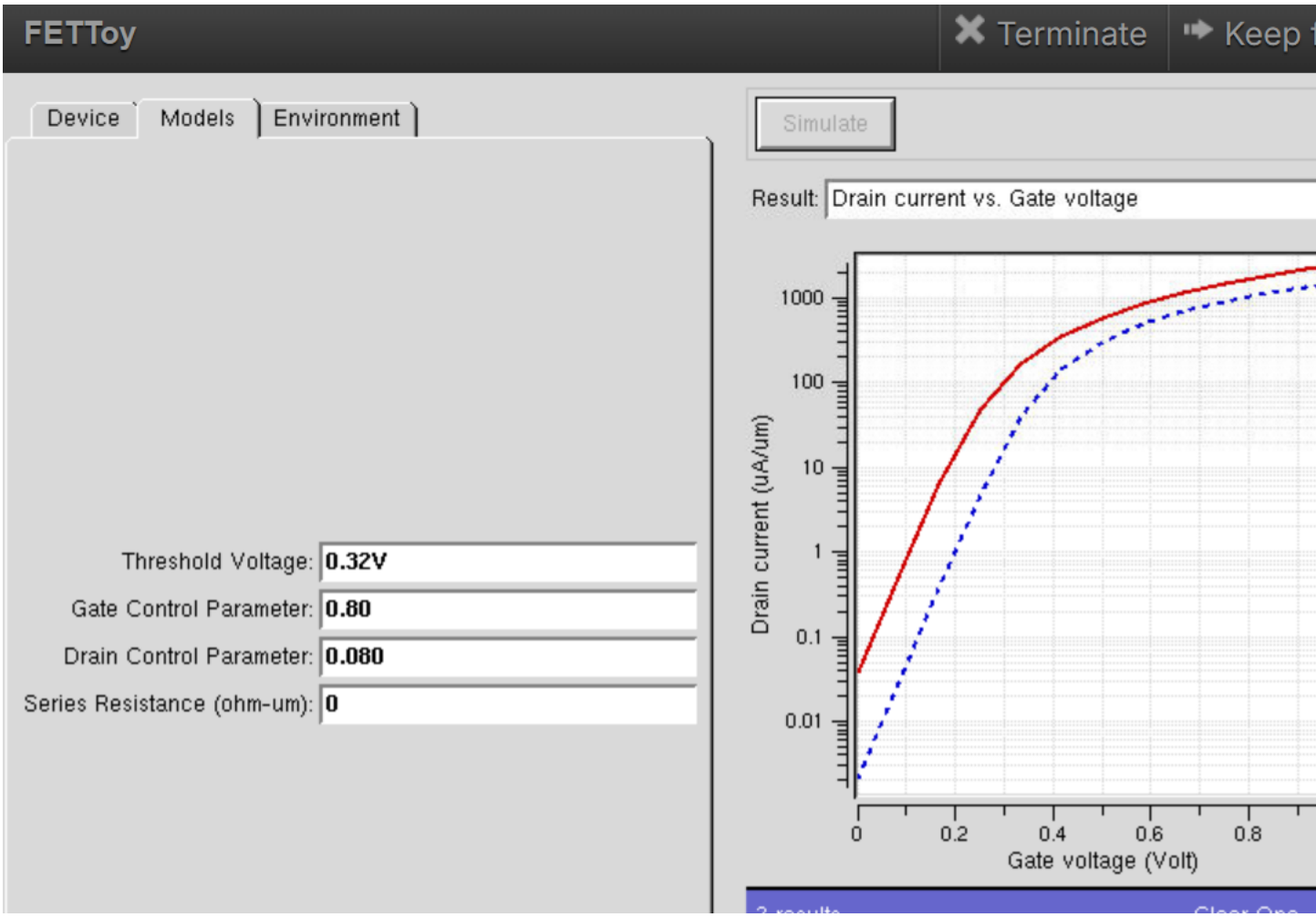
CASE 2 BASELINE

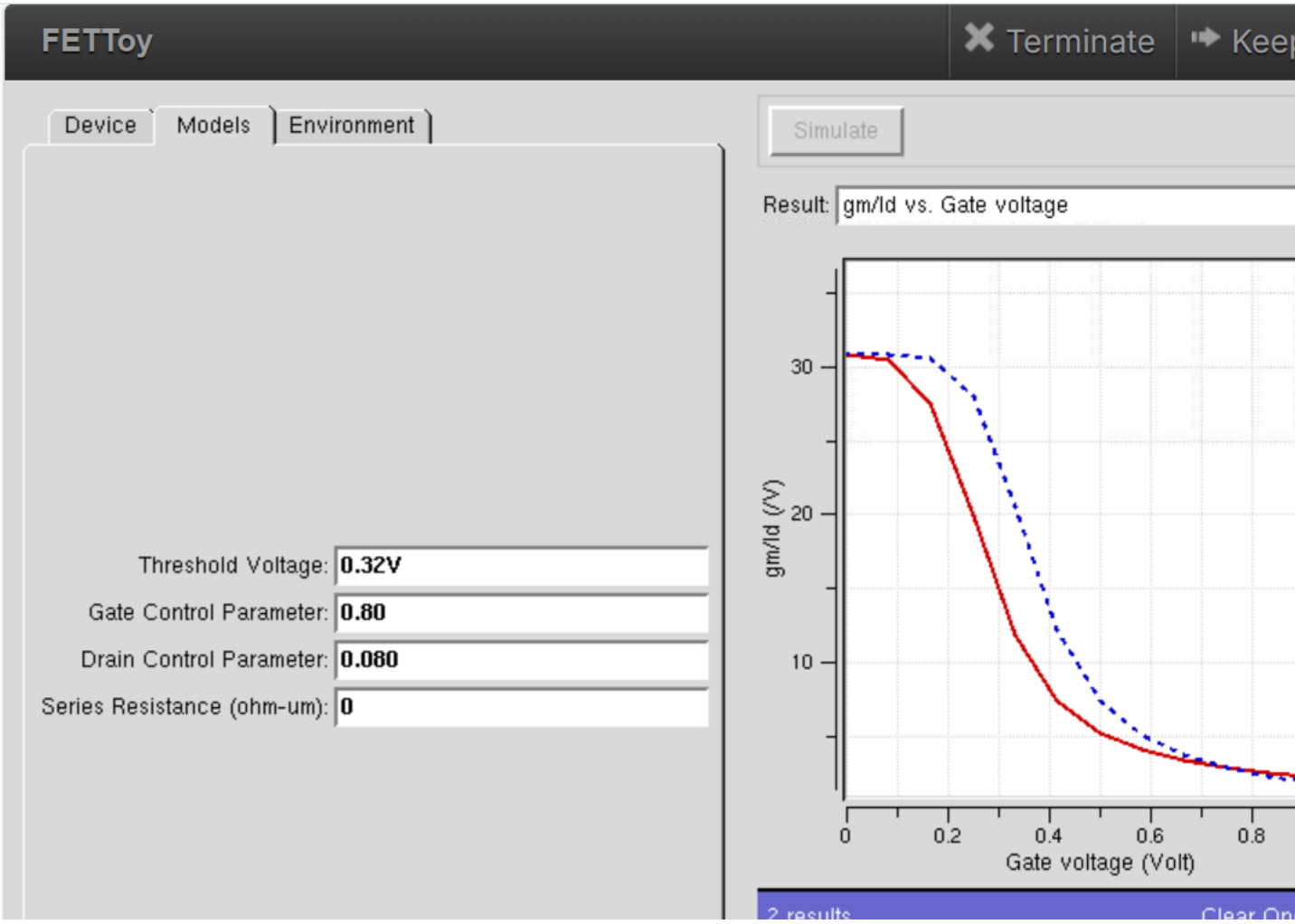



Device
Models
Environment
Model: Single-Gate MOSFET
Gate Insulator Thickness: 1.5nm
Gate Insulator Dielectric Constant: 3.9
Effective Mass Ratio: 0.19
Valley Degeneracy: 2
Floating Boundary Flag: no
Body Thickness: 10nm
Source Doping Density: 1e+20/cm3
Oxide Dielectric Constant
Gate
Electron Masses in Silicon
SiO2
Tox
Source
Tsi
Channel
Drain
Simulate
Result: Drain current vs. Gate voltage
Drain current (uA/um)
1000
100
10
1
0.1
0.01
0
0.2
0.4
0.6
0.8
1


Device
Models
Environment
Threshold Voltage: 0.32V
Gate Control Parameter: 0.88
Drain Control Parameter: 0.035
Series Resistance (ohm-um): 0
Simulate
Result: gm/Id vs. Gate voltage
gm/Id (/V)
30
20
10
0
0.2
0.4
0.6
0.8
1
Gate voltage (Volt)
3 results
Clear One
Clear All

Case3


FETToy
Terminate
Keep
Device
Models
Environment
Simulate
Result:
Drain current vs. Gate voltage
Threshold Voltage:
0.32V
Gate Control Parameter:
0.80
Drain Control Parameter:
0.080
Series Resistance (ohm-um):
0
1000
100
10
1
0.1
0.01
Drain current (uA/um)
0
0.2
0.4
0.6
0.8
Gate voltage (Volt)


FETToy
Terminate
Device
Models
Environment
Simulate
Result:
gm/Id vs. Gate voltage
Threshold Voltage:
0.32V
Gate Control Parameter:
0.80
Drain Control Parameter:
0.080
Series Resistance (ohm-um):
0
30
20
10
gm/Id (/V)
0
0.2
0.4
0.6
0.8
Gate voltage (Volt)